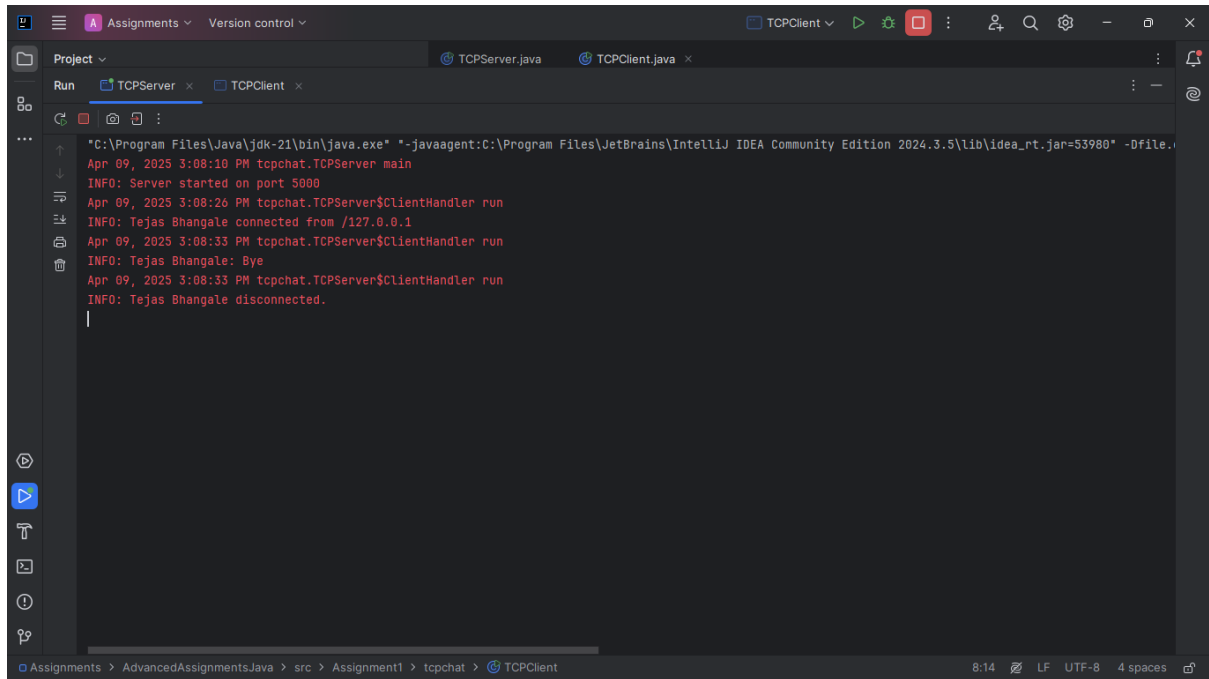
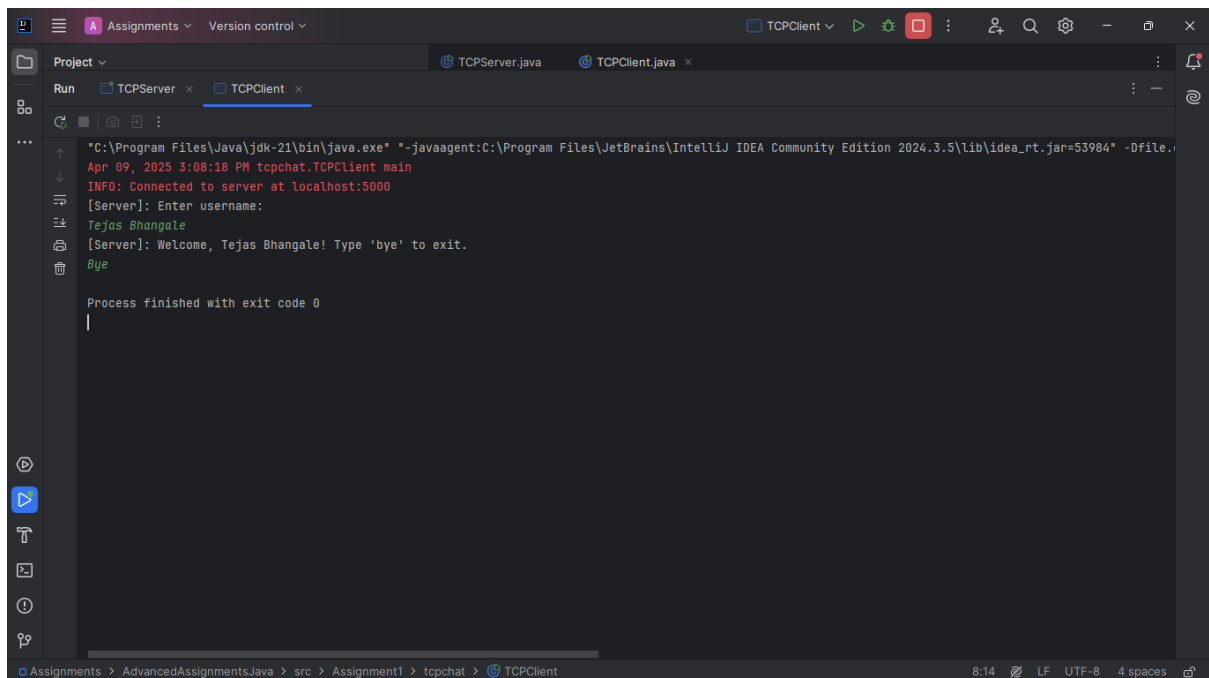


Assignment 1 - Socket Programming

BE-IT-54 – Tejas Bhargale



```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=53980" -Dfile.encoding=UTF-8
Apr 09, 2025 3:08:10 PM tcpchat.TCPServer main
INFO: Server started on port 5000
Apr 09, 2025 3:08:26 PM tcpchat.TCPServer$ClientHandler run
INFO: Tejas Bhargale connected from /127.0.0.1
Apr 09, 2025 3:08:33 PM tcpchat.TCPServer$ClientHandler run
INFO: Tejas Bhargale: Bye
Apr 09, 2025 3:08:33 PM tcpchat.TCPServer$ClientHandler run
INFO: Tejas Bhargale disconnected.
```

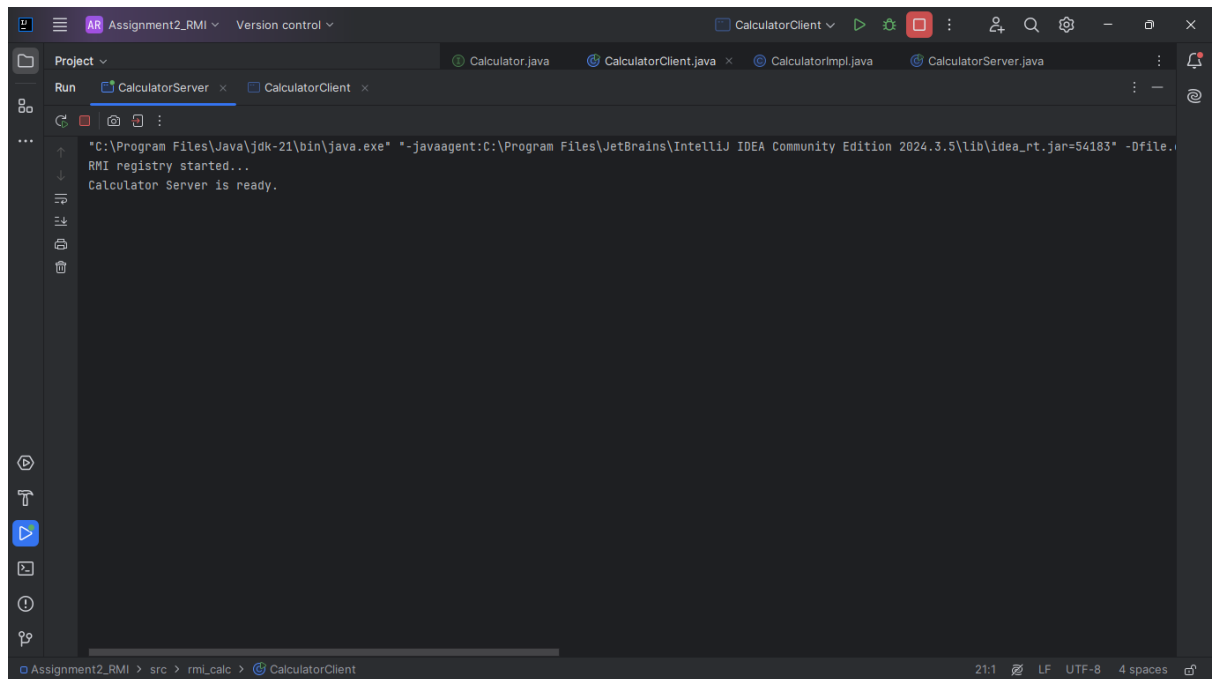


```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=53984" -Dfile.encoding=UTF-8
Apr 09, 2025 3:08:18 PM tcpchat.TCPCClient main
INFO: Connected to server at localhost:5000
[Server]: Enter username:
Tejas Bhargale
[Server]: Welcome, Tejas Bhargale! Type 'bye' to exit.
Bye

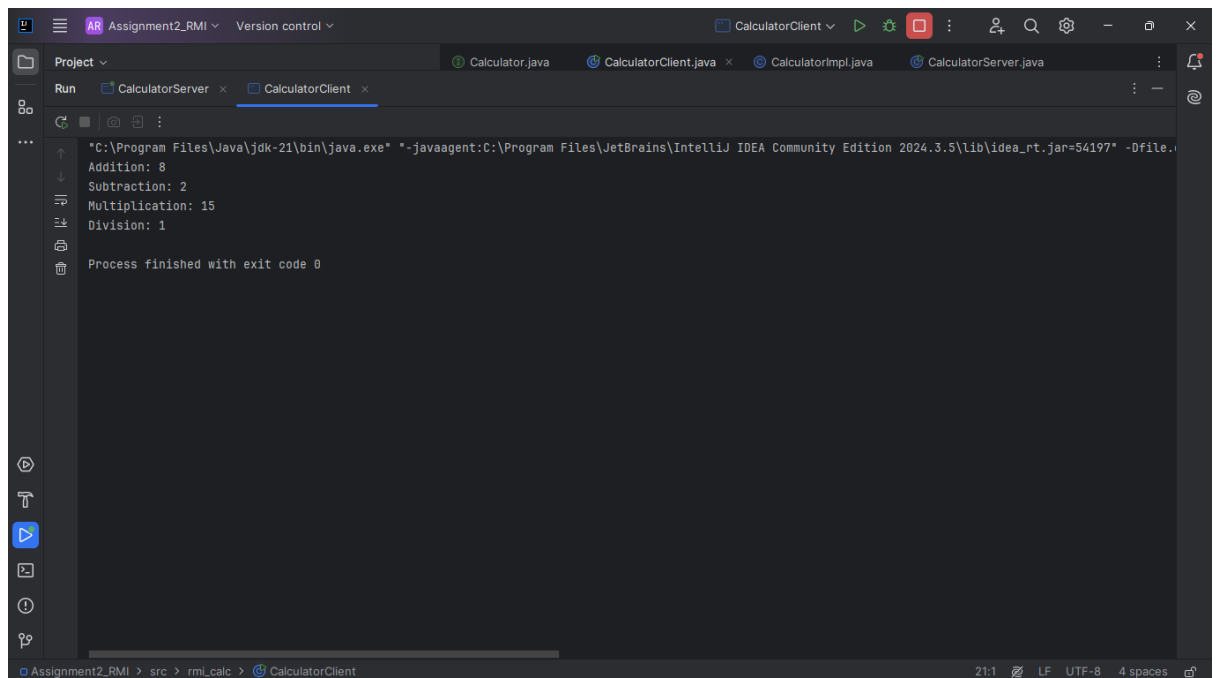
Process finished with exit code 0
```

Assignment 2 - Remote Method Invocation

BE-IT-54 – Tejas Bhangale



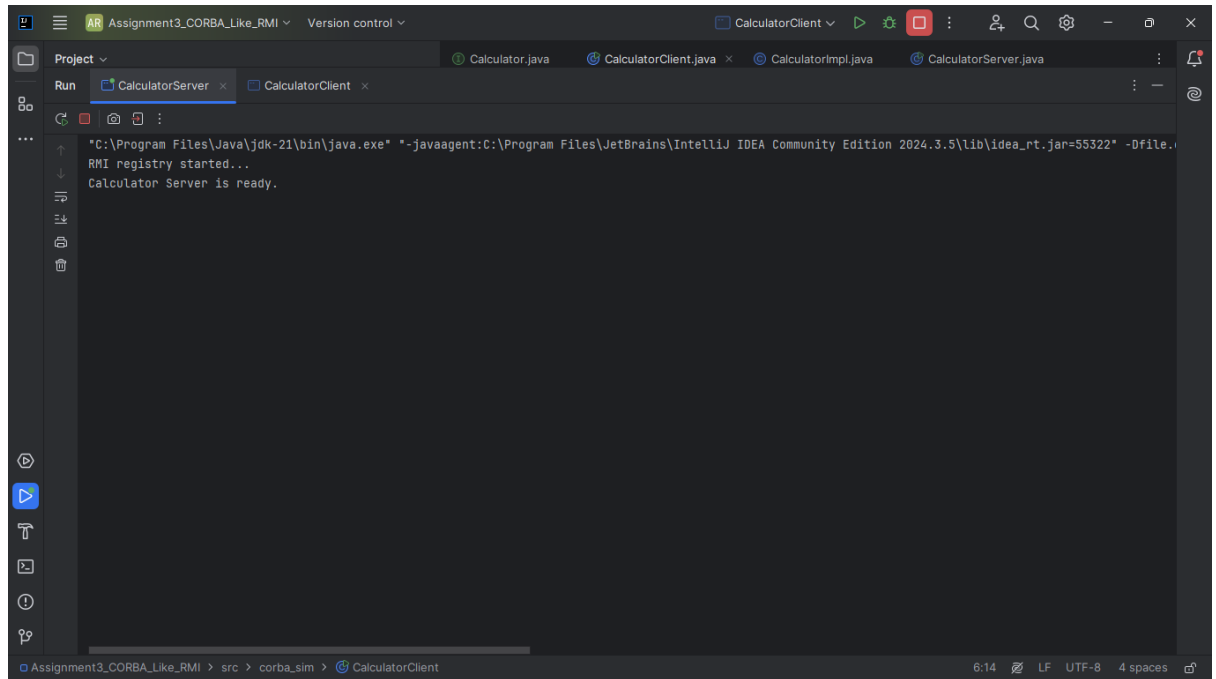
```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=54183" -Dfile.encoding=UTF-8
RMI registry started...
Calculator Server is ready.
```



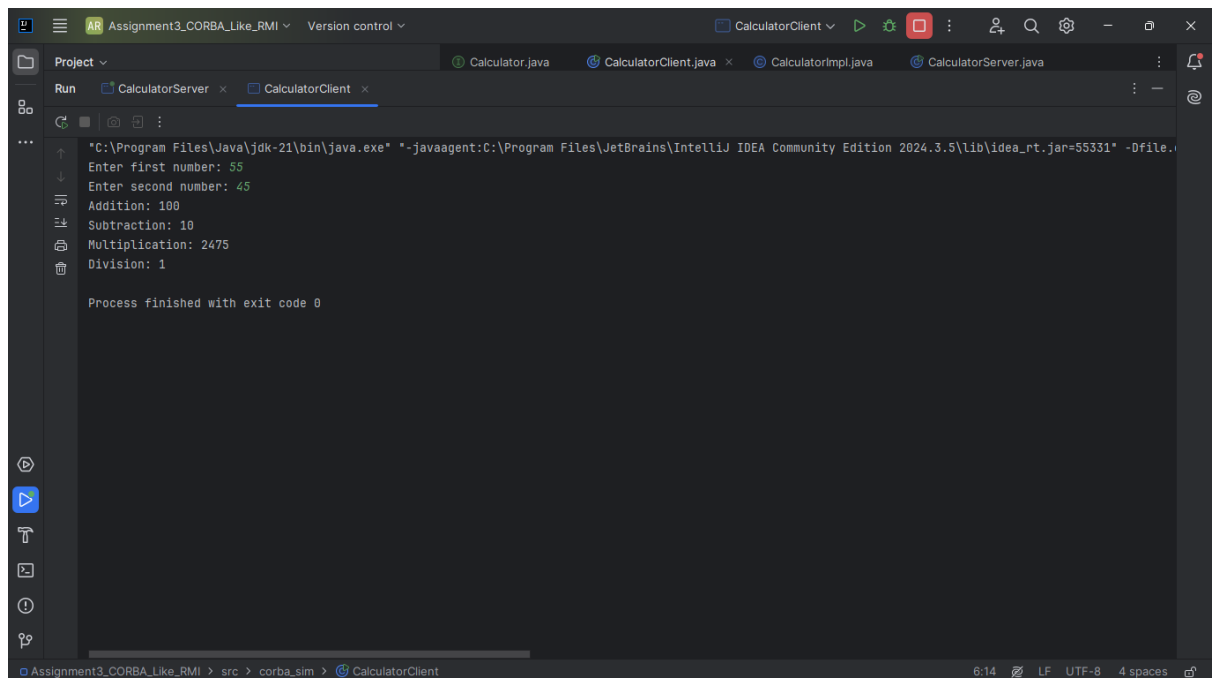
```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=54197" -Dfile.encoding=UTF-8
Addition: 8
Subtraction: 2
Multiplication: 15
Division: 1
Process finished with exit code 0
```

Assignment 3 - Common Object Request Broker Architecture (CORBA)

BE-IT-54 – Tejas Bhangale



```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=55322" -Dfile.encoding=UTF-8
RMI registry started...
Calculator Server is ready.
```

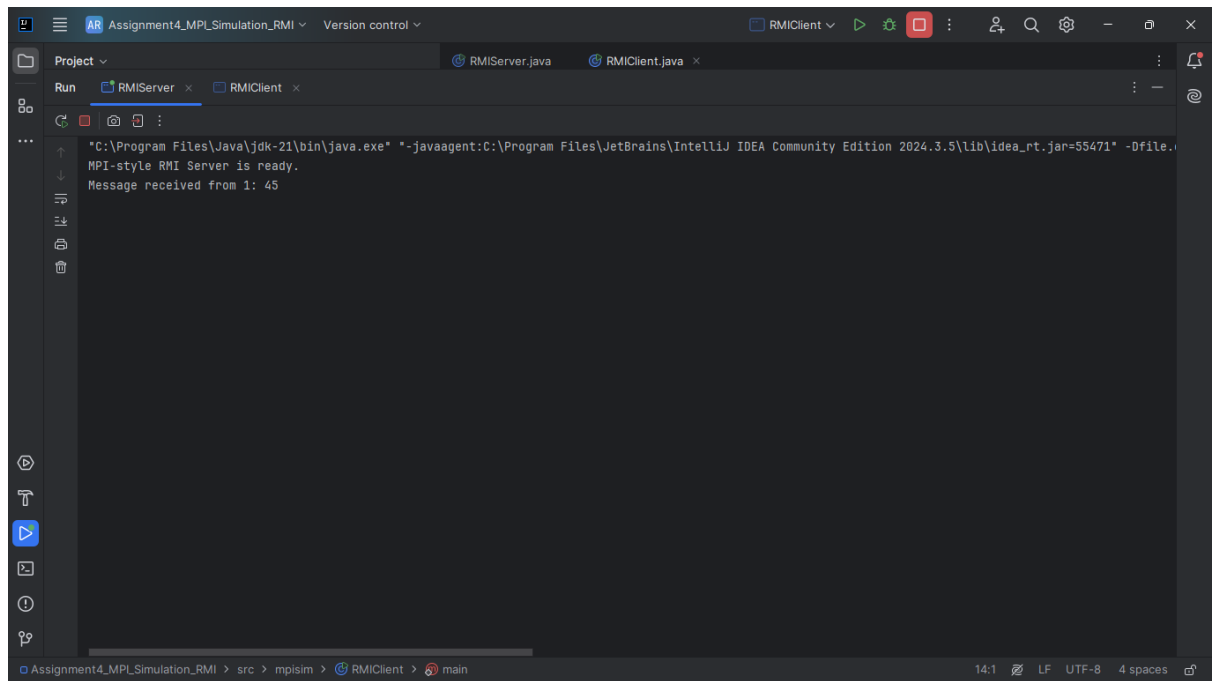


```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=55331" -Dfile.encoding=UTF-8
Enter first number: 55
Enter second number: 45
Addition: 100
Subtraction: 10
Multiplication: 2475
Division: 1

Process finished with exit code 0
```

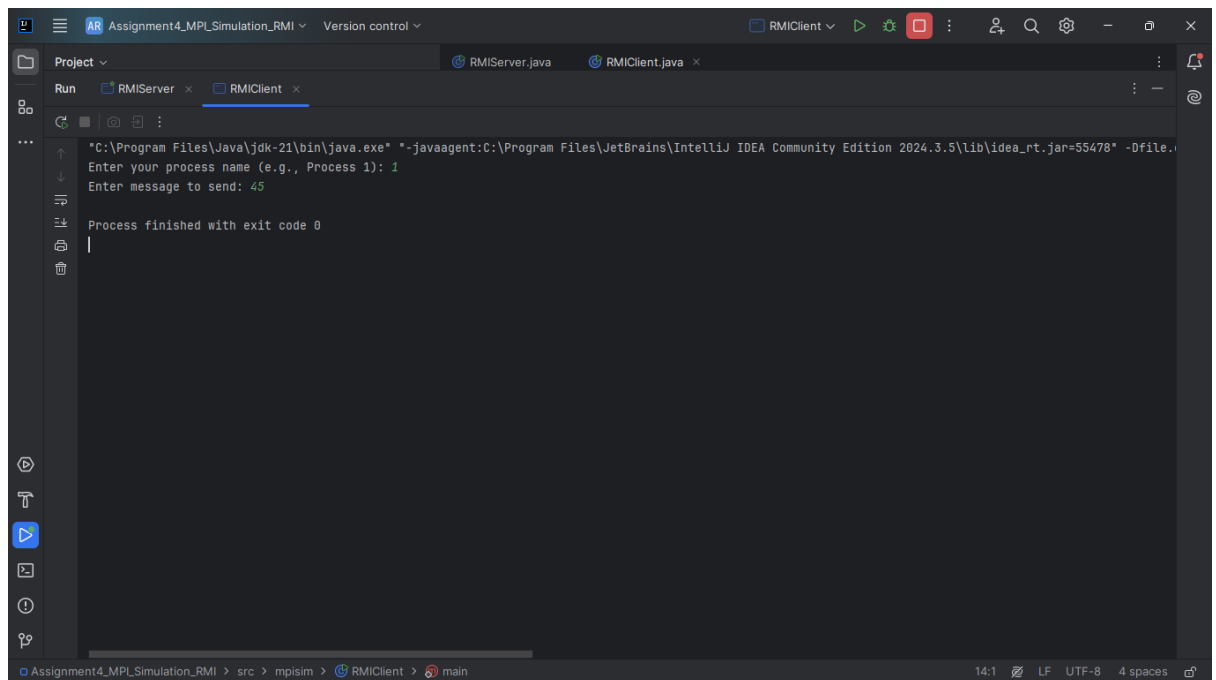
Assignment 4 - Message Passing Interface (MPI)

BE-IT-54 – Tejas Bhangale



This screenshot shows the IntelliJ IDEA interface with the 'Run' console open. The console displays the output of the RMI Server. The command executed is: `"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=55471" -Dfile.encoding=UTF-8`. The output shows: `MPI-style RMI Server is ready.` and `Message received from 1: 45`. The project structure at the bottom indicates the file `RMIClient.java` is being run.

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=55471" -Dfile.encoding=UTF-8
MPI-style RMI Server is ready.
Message received from 1: 45
```

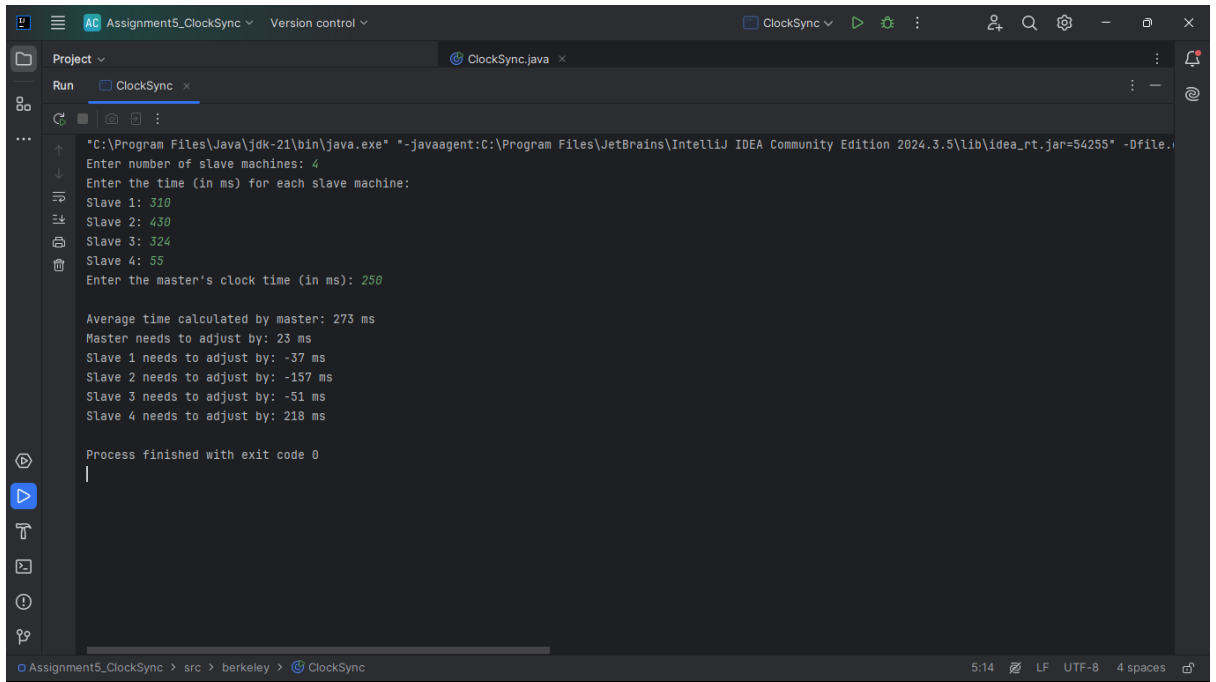


This screenshot shows the IntelliJ IDEA interface with the 'Run' console open. The console displays the output of the RMI Client. The command executed is: `"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=55478" -Dfile.encoding=UTF-8`. The output shows: `Enter your process name (e.g., Process 1): 1`, `Enter message to send: 45`, and `Process finished with exit code 0`. The project structure at the bottom indicates the file `RMIClient.java` is being run.

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=55478" -Dfile.encoding=UTF-8
Enter your process name (e.g., Process 1): 1
Enter message to send: 45
Process finished with exit code 0
```

Assignment 5 - Clock Synchronization

BE-IT-54 – Tejas Bhangale



The screenshot shows an IDE window titled "Assignment5_ClockSync" with a "Run" tab active. The console output displays the execution of a Java program that synchronizes four slave machines with a master. The program prompts for the number of slave machines (4), the time for each slave machine (310, 430, 324, 55 ms), and the master's clock time (250 ms). It then calculates the average time (273 ms) and the adjustment needed for each slave machine.

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=54255" -Dfile.encoding=UTF-8
Enter number of slave machines: 4
Enter the time (in ms) for each slave machine:
Slave 1: 310
Slave 2: 430
Slave 3: 324
Slave 4: 55
Enter the master's clock time (in ms): 250

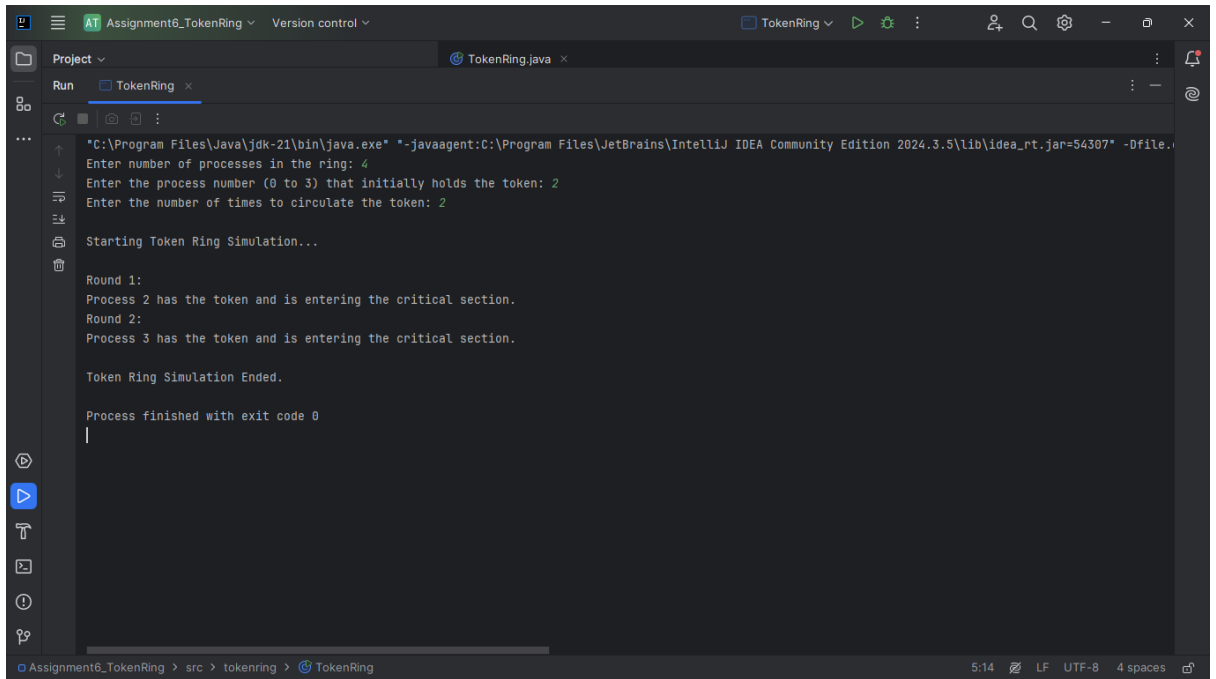
Average time calculated by master: 273 ms
Master needs to adjust by: 23 ms
Slave 1 needs to adjust by: -37 ms
Slave 2 needs to adjust by: -157 ms
Slave 3 needs to adjust by: -51 ms
Slave 4 needs to adjust by: 218 ms

Process finished with exit code 0
```

Assignment5_ClockSync > src > Berkeley > ClockSync 5:14 LF UTF-8 4 spaces

Assignment 6 - Mutual Exclusion

BE-IT-54 – Tejas Bhangale



The screenshot shows an IDE window titled "Assignment6_TokenRing" with a "Run" tab active. The console output displays the execution of a Java program named "TokenRing.java". The program prompts the user for the number of processes (4), the initial token holder (2), and the number of circulations (2). It then simulates a token ring where Process 2 enters the critical section in Round 1, and Process 3 enters in Round 2. The simulation ends with the message "Token Ring Simulation Ended." and "Process finished with exit code 0".

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=54307" -Dfile.encoding=UTF-8
Enter number of processes in the ring: 4
Enter the process number (0 to 3) that initially holds the token: 2
Enter the number of times to circulate the token: 2

Starting Token Ring Simulation...

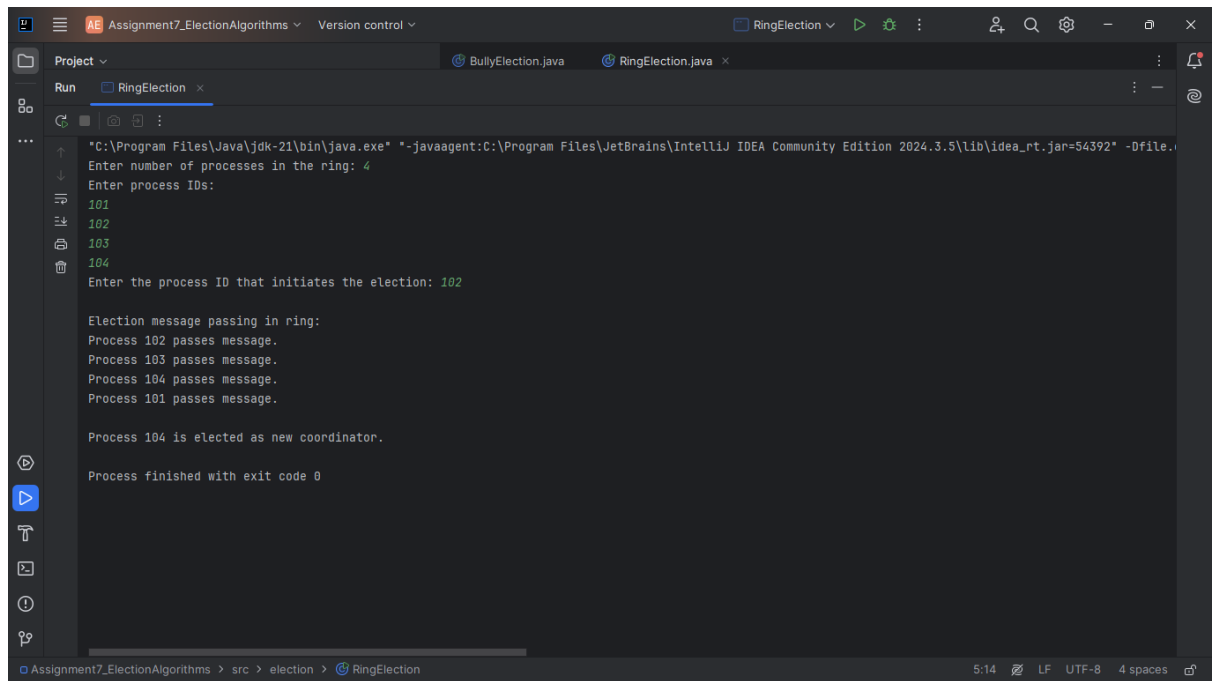
Round 1:
Process 2 has the token and is entering the critical section.
Round 2:
Process 3 has the token and is entering the critical section.

Token Ring Simulation Ended.

Process finished with exit code 0
```

Assignment 7 - Election Algorithms

BE-IT-54 – Tejas Bhangale

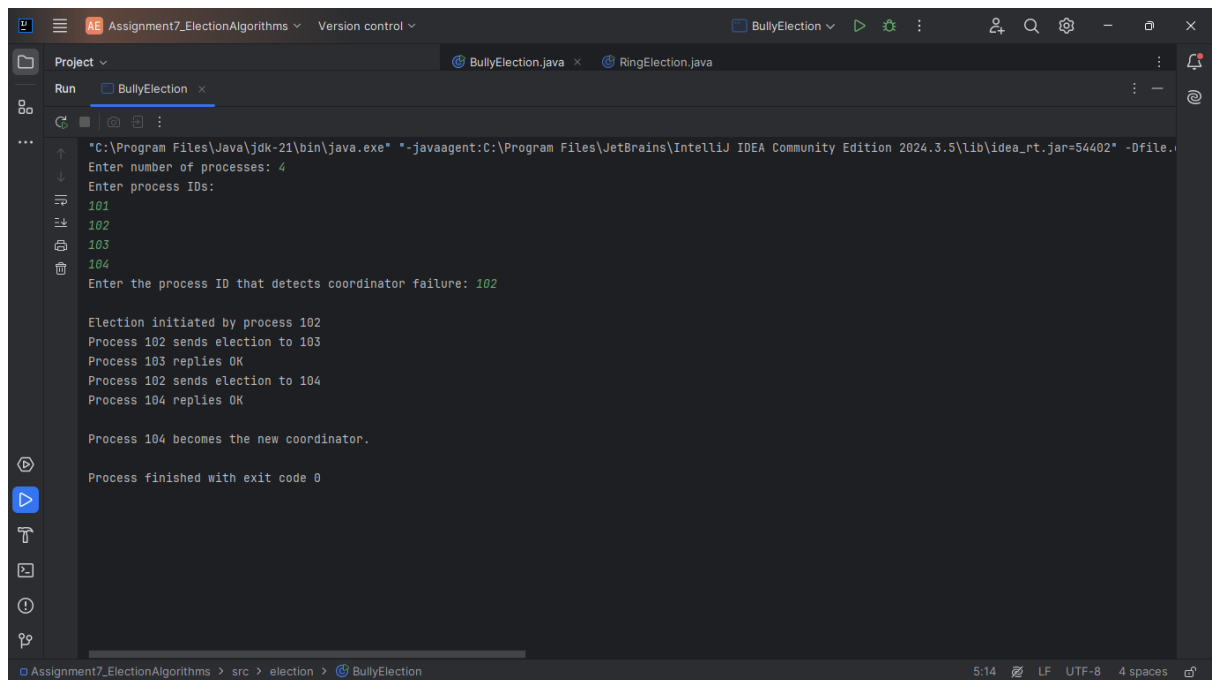


```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=54392" -Dfile.encoding=UTF-8
Enter number of processes in the ring: 4
Enter process IDs:
101
102
103
104
Enter the process ID that initiates the election: 102

Election message passing in ring:
Process 102 passes message.
Process 103 passes message.
Process 104 passes message.
Process 101 passes message.

Process 104 is elected as new coordinator.

Process finished with exit code 0
```



```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=54402" -Dfile.encoding=UTF-8
Enter number of processes: 4
Enter process IDs:
101
102
103
104
Enter the process ID that detects coordinator failure: 102

Election initiated by process 102
Process 102 sends election to 103
Process 103 replies OK
Process 102 sends election to 104
Process 104 replies OK

Process 104 becomes the new coordinator.

Process finished with exit code 0
```

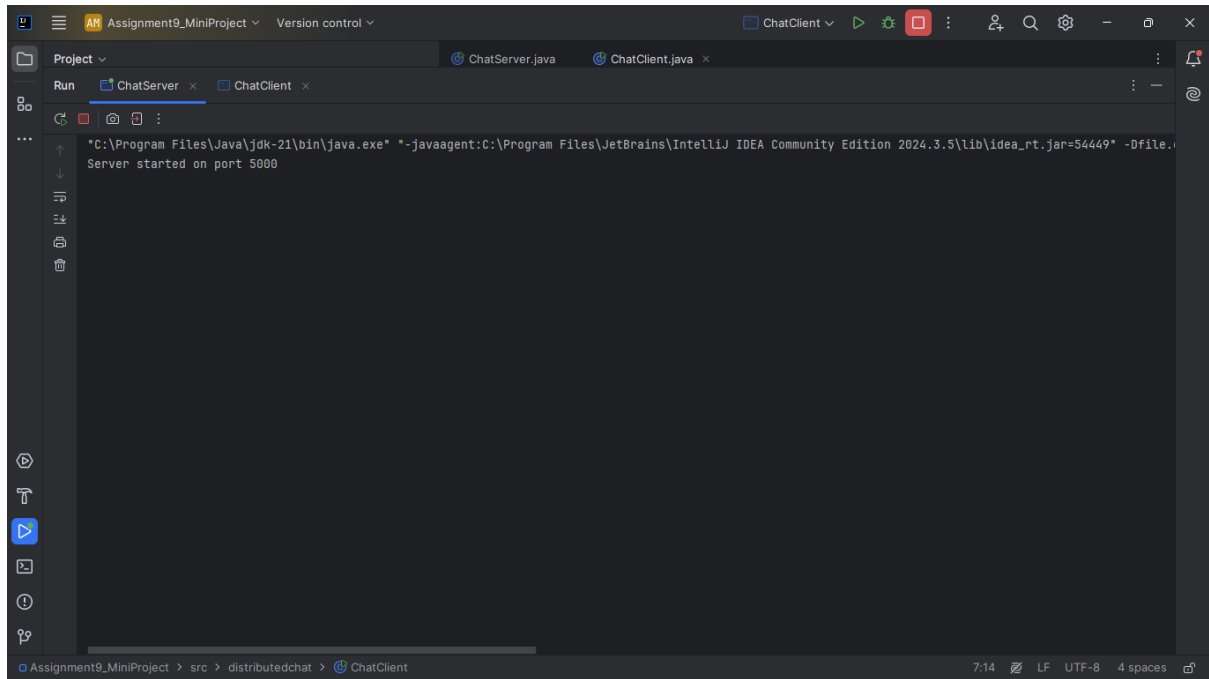
BE-IT-54 – Tejas Bhangale

Compiling localhost on web browser -



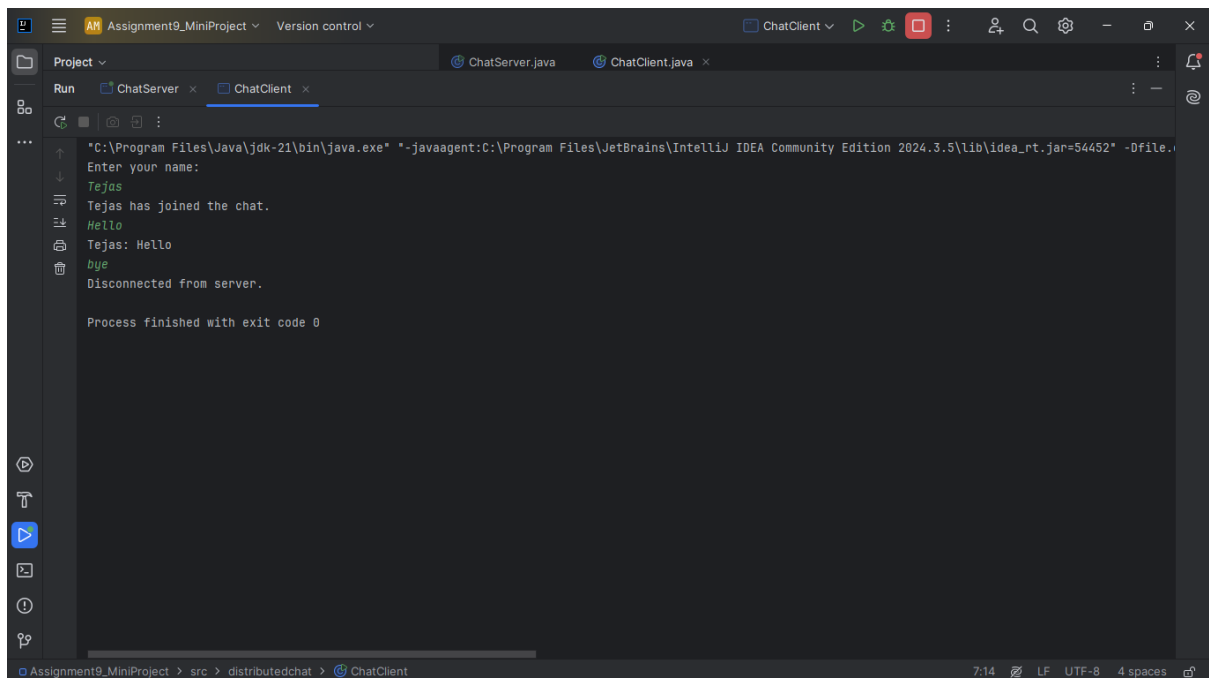
Assignment 9 - Mini Project (Distributed Chat Application)

BE-IT-54 – Tejas Bhangale



The screenshot shows the IntelliJ IDEA interface with the 'Run' tab selected. The console output for 'ChatServer.java' is visible, showing the command used to run the application and the message 'Server started on port 5000'.

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=54449" -Dfile.encoding=UTF-8
Server started on port 5000
```



The screenshot shows the IntelliJ IDEA interface with the 'Run' tab selected. The console output for 'ChatClient.java' is visible, showing the command used to run the application and the sequence of messages exchanged between the client and the server.

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.3.5\lib\idea_rt.jar=54452" -Dfile.encoding=UTF-8
Enter your name:
Tejas
Tejas has joined the chat.
Hello
Tejas: Hello
bye
Disconnected from server.

Process finished with exit code 0
```